1.1 STEPS IN RESEARCH

Researchers, or experimenters, use a series of orderly and systematic steps to plan, conduct, interpret and report on the research they undertake. Seven steps are commonly used in conducting psychological research. Read through the steps below to gain a holistic understanding of how the research process progresses.

Step 1: Identify the research problem

The first step in psychological research is to identify an area that needs investigating. When a psychological researcher has a topic of interest, they must conduct a *literature search*. This involves finding and reading relevant research articles, results and journals on the topic. Once they have completed the literature search, the researcher must develop a testable research question or problem. For example, a researcher may be interested in investigating the causes of, or influences on, eating disorders in adolescent females. The researcher would do a literature search on the broad topic, then identify a research question, such as: *Is the prevalence of eating disorders in adolescent girls affected by whether they have had a relative who has suffered from an eating disorder currently or in the past?*

Step 2: Formulate a hypothesis

The next step in psychological research is to formulate a **hypothesis**. A hypothesis is a testable prediction about the relationship between two or more variables (events or characteristics). In other words, a hypothesis is an educated guess about the results of the experiment; for example, *It is hypothesised that adolescent girls who have had a relative who has suffered from an eating disorder are more likely to have an eating disorder than those with no relatives who have suffered from an eating disorder.*

A hypothesis is based on the information gathered in the literature search; it is not just a guess made without any prior knowledge or investigation. The hypothesis is developed before the research is conducted and is written as a very specific statement.

Step 3: Design the method

The third step is to design the method that will be used to undertake the research. The research method determines how the researcher will test the hypothesis. When designing the method the researcher must determine who will participate in the study and how to sample them, how many participants are required for the study, what the participants will do and under what conditions, and what will be measured. This is also the stage in which researchers try to minimise any unwanted variables that may affect the research when it is being conducted.

Step 4: Collect the data

The fourth step is to collect the data. Data-collection procedures include controlled experiments, naturalistic observations, surveys, interviews and case studies. To measure participants' performance, opinions or results, a researcher can use a variety of data-collection techniques, including questionnaires, direct observation, psychological tests or examination of files and documents.

Step 5: Analyse the data

Data analysis involves organising, summarising and representing the raw data in a coherent and logical manner. **Raw data** are the actual data collected from a study. A research investigation will often result in large amounts of raw data that are then made sense of through statistical procedures and summaries. **Descriptive statistics** are used to describe, summarise and organise data. Examples of descriptive statistics include graphs; percentages; tables; the mean, median and mode; and frequency distributions.

Step 6: Interpret the results

Once the data have been summarised and organised, they must be interpreted. Interpreting results involves forming conclusions about what the data shows. **Inferential statistics** are statistics that allow you to make inferences and conclusions about the data, and are often used to interpret results. Such statistics allow us to explain the significance of the data in light of the variables manipulated. At this stage, a **conclusion** may be drawn which is a decision or judgement about the meaningfulness of the results of a study.

Step 7: Report the findings

There is no point in conducting psychological research if you cannot share it with other psychologists and researchers. Therefore, the final step in psychological research is to report the data to others. After conducting a study, the researcher usually prepares a report that is either submitted to a journal, periodical, incorporated into a thesis or is presented to other psychologists. Once a research report is published, other researchers use it in *their* literature searches and further investigations. Publication also enables the general public to benefit from research findings.

ISBN 9780170365642

1 Fill in the flow chart to identify the seven steps in psychological research. Next to each step, give an example of what *may* be done or found in each step. The research problem has been identified for you.

Step 1 → Does listening to music while studying improve the retention of the material you are studying?

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Step 2 → It is hypothesised that . . .

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Step 3 → Participants will . . .

↓

Step 4 → Data will be collected by . . .

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Step 5 → The raw data show that . . .

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Step 6 → The findings suggest that . . .

↓

Step 7

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1.2 INDEPENDENT AND DEPENDENT VARIABLES

An **experiment** is a study that investigates a cause-and-effect relationship between two or more variables; that is, whether a change in one thing has an impact on another. A **variable** is any condition that can change. For example, if research involved testing the effectiveness of a new memory drug on participants' retention of a list of nonsense words, two variables would be involved when conducting this experiment: whether or not participants are given the drug, and the retention of the nonsense words. The first variable would be manipulated by the experimenter: whether the participants are exposed to the experimental condition or not. The second variable is dependent on the participants, as it is their results or responses.

The independent variable

An **independent variable (IV)** is a condition that an experimenter systematically manipulates, changes or varies in order to gauge its effect on another variable. The independent variable is that which is different between the groups involved in the study. If we have two groups of participants in an experiment, one group would be exposed to the independent variable, or experimental condition, and the other would not. One group may be sleep deprived for 24 hours and the other sleep normally through the night; or, one group might eat five meals a day and the other group three. The researcher can then see the difference between the results of each group and potentially make inferences about whether the IV can be said to have caused that difference. The results in this case would be known as the *dependent variable*. An experiment measures the effect that the IV has on the dependent variable. Put simply, the IV is identified by looking at what is different between the experimental conditions.

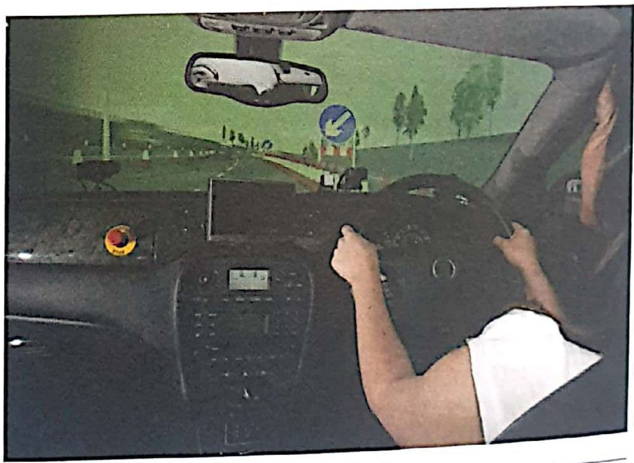
The dependent variable

The **dependent variable (DV)** is that which is measured in an experiment. Dependent variables measure the effects that the manipulation of the IV has had on behaviour. Such effects are often revealed by measures of performance, such as test scores or number of goals scored. A DV 'depends' on another variable, the IV, to determine whether it changes and by how much. To find the DV, look for what is being measured.

Operationalising variables

Once you have learnt to identify IVs and DVs in experiments, it is important to that you learn to **operationalise** them. An operationalised variable explains what each variable is and how it will be measured. For example, a researcher examining the effects of a new treatment therapy for paranoid schizophrenia might divide a sample into two groups. He could ensure that participants in one group take the new drug every day for a week and that participants in the other group have no treatment at all. The experimenter is manipulating this condition. He could then measure the number of hallucinations the patients in each group reportedly experience over the week. In this scenario the operationalised IV is the presence (or not) of the new drug and the operationalised DV is the number of hallucinations reportedly experienced in a one-week period. This experiment allows the researcher to see whether the presence of the drug (IV) affects the experience of hallucinations (DV).

As another example, a study into the effects of alcohol on driving may involve one group of participants who drink five alcoholic beverages in an hour and one group of participants who drink five glasses of water in an hour. The participants may then be asked to navigate an obstacle course on a driving simulator, in which they must avoid hitting the witches' hats. In this experiment, the experimenter is manipulating whether the participants drink alcohol or water, and the results will be measured by the number of witches' hats they hit. Therefore, the operationalised IV is the type of drink the participants drink and the operationalised DV is the number of witches' hats the participants hit when operating a driving simulator. This experiment allows the researcher to see whether the presence of alcohol (IV) affects driving ability (DV).



Alamy/Tom Wood

The dependent variable can be operationalised for example, by counting the number of witches, hats hit when using a driving simulator.

As you can see from the examples given, the research itself involves assessing the effect of the IV on the DV. It is important to operationalise both of these variables so that we can see in detail what is actually manipulated and to what degree, and what is actually measured and how it is measured.

ISBN 9780170365642

CHECK YOUR UNDERSTANDING

1 Identify the operationalised independent and dependent variables in each experiment.

a Professor Tallent is examining the effect of hunger as motivation in rats trying to run through a maze. She uses two groups of rats, and times how long it takes members of each group to successfully complete the maze to find a piece of cheese. Professor Tallent feeds the rats in one group before they enter the maze, but does not feed the rats in the other group until they have completed the maze.

i Operationalised IV: ______

ii Operationalised DV: ______

b Emma has always been told that she makes excellent muffins, but she wants to know which of her two favourite flavours she makes is the best. For two months (January and February), Emma makes muffins and distributes them to her friends and family. She makes the same number of muffins each month and distributes them to the same people, but for the month of January she makes only blueberry muffins and for the month of February she makes only chocolate muffins. Emma counts the number of compliments she receives on her muffins over the two months.

i Operationalised IV: ______

ii Operationalised DV: ______

c Luli was in charge of a large corporation that taught people second languages for travel purposes. She had equal numbers of students in two six-week programs: one where students learned by hearing and repeating the foreign words and another where students learned by reading and writing the foreign words. Luli decided to test which method was most effective by giving the students of each program a test at the end of the six weeks.

i Operationalised IV: ______

ii Operationalised DV: ______

2 Parents often tell children old wives' tales and stories to discourage them from displaying undesirable behaviours. Design an experiment to test the following well-known statements. Remember to operationalise your independent and dependent variables.

a Sitting too close to the television is detrimental to your vision.

Experiment design: ______

Operationalised IV: ______

Operationalised DV: ______

b Eating bread crusts makes your hair curly.

Experiment design: ______

Operationalised IV: ______

Operationalised DV: ______

1.3 FORMULATING A HYPOTHESIS

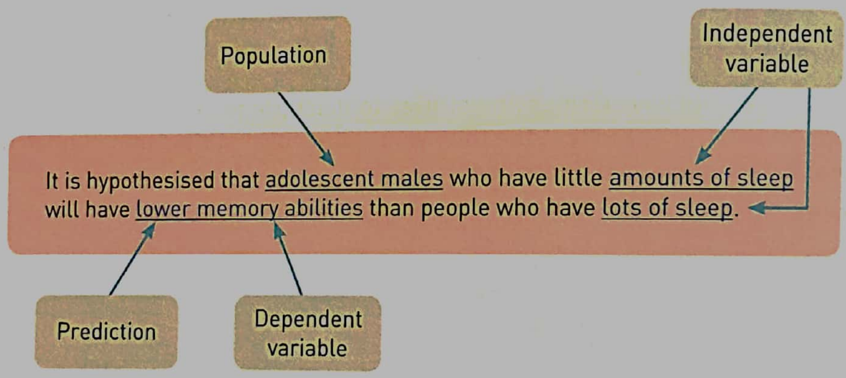
Once a researcher has identified the variables at play in investigating a research question, they need to make an educated guess about the relationship between these two variables. A hypothesis is a statement, or testable prediction, about the likely outcome of an experiment.

A **research hypothesis** is a general prediction about the direction of interaction between the IV and the DV and also includes the population from which the sample is drawn. That is, when the IV is manipulated, does the DV increase or decrease? For example, you might predict that short periods of sleep lead to decreased memory ability in adolescent males. This is obviously a very general statement; it does not tell us how much sleep or what *types* of memory ability are being referred to, or how memory ability might be being measured.

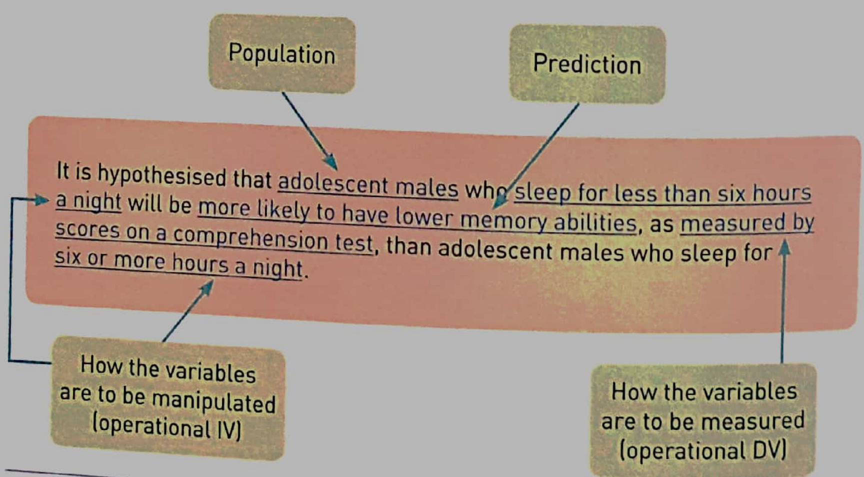
An **operational hypothesis**, on the other hand, is a testable prediction that explains exactly how the variables will be measured and manipulated, as well as the population from which the sample is drawn. Thus, an operational hypothesis is a workable, testable and repeatable hypothesis.

It is important that any hypothesis you write includes four elements:

1. a testable prediction about the direction of interaction between variables (i.e. higher, lower, increased, decreased, etc.)
2. the population from which the sample is drawn
3. the independent variable (that which is manipulated); operationalised, if it is an operational hypothesis
4. the dependent variable (that which is measured); operationalised, if it is an operational hypothesis.



A research hypothesis



An operational hypothesis

CHECK YOUR UNDERSTANDING

For each of the following scenarios, identify the operationalised independent variable and dependent variable. Formulate a research hypothesis and an operational hypothesis for each.

1 Max is conducting research to investigate whether sleep quality is improved by meditating. He investigates a group of 40-year-old males, where half of the group practises meditation for 10 minutes before going to bed every night for one week, and the other half does not. Upon waking each morning for that week, the participants in both groups are to rate their quality of sleep on a scale from 1 to 10. An average of each participant's seven ratings is taken.

Operationalised IV: ____________________

Operationalised DV: ____________________

Research hypothesis: ____________________

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Operational hypothesis: ____________________

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2 When a Year 12 Psychology teacher taught her lessons in the first half of the year, she wrote notes on the board for her class to copy. She was disappointed in the results obtained by her class in the mid-year exam, and wanted to investigate whether presenting her notes in a different way would impact on her students' results in the end-of-year exam. In the second half of the year, rather than write her notes on the board, she presented her notes in PowerPoint, and added lots of visual cues.

Operationalised IV: ____________________

Operationalised DV: ____________________

Research hypothesis: ____________________

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Operational hypothesis: ____________________

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3 Dr Sing is interested in assessing whether Victorians' memory abilities decrease with age. She places an advertisement in the newspaper calling for 20- to 30-year-olds and 50- to 60-year-olds to take part in her study. She then exposes participants in both age groups to a series of memory-related tests and compares the performances of the two groups using average scores obtained from the tests.

Operationalised IV: ____________________

Operationalised DV: ____________________

Research hypothesis: ____________________

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Operational hypothesis: ____________________

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